

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2021

Bachelor in Computer Applications

Course Title: **Digital Logic**

Code No: **CACS103** Semester: **I**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. Subtract: 453.35 - 321.17 using both 9's and 10's complement.
3. Simplify given Boolean function in both SOP and POS using k-map where d represents don't care condition.
 $F(A, B, C, D) = \prod (2, 3, 4, 5, 7, 9, 10)$ and $\prod d (0, 6, 11)$
4. Define combinational logic circuit. Design a full adder with its truth table, logic diagram and Boolean expression.
5. Define Decoder. Design BCD to Decimal decoder with its block diagram, truth table, circuit diagram and mathematical expression.
6. Explain basic gates and universal gate. Realize NOR gate as a universal gate.
7. Explain clocked JK flip flop with logic diagram, truth table, characteristic table and excitation table.
8. Differentiate between synchronous and asynchronous counter. Design 3-bit up ripple counter.

Group C

Attempt any TWO questions[2x10=20]

9. Explain PLA with its block diagram. Design a combinational circuit using a ROM that accepts a 3-bit number and generates an output binary number equal to the square of the input number with circuit diagram, truth table and block diagram.
10. What is counter? Design a mod 7 synchronous counter with state table, state diagram, circuit diagram.
11. Define multiplexer. Explain 4:1 multiplexer with its block diagram, truth table and logic diagram. Mention how a 4:1 multiplexer works like lower order multiplexer.